



Birzeit University

Department of Electrical & Computer Engineering

First Semester, 2024/2025

ENCS5343 Computer Vision

Assignment#2

**Arabic Handwritten Text Identification Using Local Feature Extraction
Techniques**

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Section: Sec1

Date: 15/12/2024

Introduction

This assignment looks at how local feature extraction techniques can be used to identify Arabic handwritten text. It focuses on comparing two popular algorithms: Scale-Invariant Feature Transform (SIFT) and Oriented FAST and Rotated BRIEF (ORB). The AHAWP dataset, which contains a variety of handwritten Arabic word samples, was divided into training and testing sets. The system works by taking a test image, matching it with the training set, and identifying the writer of the text. The accuracy of both algorithms is then evaluated to see how well they perform. This assignment highlights the key role of feature extraction in computer vision, especially in identifying consistent key points even with changes in scale, rotation, or illumination. By testing and comparing the results, the report provides insights into the strengths and weaknesses of SIFT and ORB for recognizing handwritten text.

The methods used

The methods used revolve around two popular feature extraction and matching techniques: **SIFT** (Scale-Invariant Feature Transform) and **ORB** (Oriented FAST and Rotated BRIEF). **SIFT** is a robust method for detecting and describing local features in images, which is invariant to scale, rotation, and partially invariant to affine transformations. It works by identifying key points in an image and computing descriptors based on the gradient information around those key points, making it highly effective for object recognition and image matching tasks. However, **SIFT** is computationally expensive and can be slow, especially for large datasets.

On the other hand, **ORB** is a more efficient alternative that is based on the **FAST** key point detector and **BRIEF** descriptor. **ORB** is designed to be faster than **SIFT**, making it suitable for real-time applications. It achieves rotation invariance using the orientation of key points and is less computationally demanding compared to **SIFT**. While **ORB** may not be as robust as **SIFT** in some cases, it strikes a balance between performance and speed, making it suitable for applications requiring real-time processing. We evaluate these methods in terms of execution time, robustness under various transformations (such as scale, rotation, brightness, and noise), and key point detection, offering insights into the strengths and weaknesses of each method.

Experiments

Step 1:

Builds a machine learning pipeline to classify isolated spoken words represented as images. It downloads a dataset, preprocesses images by resizing and converting them to grayscale, and extracts features using SIFT and ORB methods. These features are converted into fixed-length histograms using k-means clustering to create a Bag-of-Words (BoW) representation. Two SVM classifiers are trained—one for SIFT and one for ORB features—to predict user labels based on the visual patterns in the images. The models are evaluated on a test set by comparing predictions to true labels.

Step 2:

Evaluates the keypoint matching accuracy for SIFT and ORB feature extraction methods on a test dataset. For each test image, keypoints and descriptors are extracted using SIFT or ORB, and a Bag-of-Words histogram is generated. The SVM classifier predicts the user label based on this histogram. If the prediction is correct, the total number of keypoints for the image is added to the count of matched keypoints. Finally, calculates the accuracy as the ratio of matched keypoints to total keypoints for all test images, providing a measure of how effectively the extracted keypoints contribute to correct predictions.

Step 3:

Measures the execution time for feature extraction and classification using the SIFT and ORB methods on a test dataset.

Step 4:

Evaluates the robustness of SIFT and ORB feature extraction methods against various image transformations, including scaling, rotation, brightness adjustments, and noise. The function `apply_transformations` modifies images according to the specified parameters, such as scaling down/up, rotating, changing brightness, or adding Gaussian noise. The `evaluate_robustness` function applies these transformations to a set of test images, extracts features, computes Bag-of-Words histograms, and uses a trained SVM to predict labels. The accuracy of the predictions is then calculated and printed for both SIFT and ORB methods under each transformation.

Step 5:

Compares the number of key points detected by SIFT and ORB across a dataset. The `count_keypoints` function processes each image in the test set, extracts features using the specified method (SIFT or ORB), and counts the number of detected key points. The average number of key points detected by both methods is then calculated and displayed. Finally, a histogram is plotted to visualize the distribution of key points detected by SIFT and ORB, with each method shown in different colors for clarity.

Results and visualizations

Result from step 1:

The SIFT-based SVM achieved a low accuracy of **15%**, while the ORB-based SVM performed slightly worse at **13%**. Both models struggled to classify the test images correctly, frequently mispredicting user IDs. This poor performance is likely due to insufficient or noisy data, the large number of user classes, and the limitations of the Bag-of-Words approach in capturing complex spatial patterns. These results suggest a need for improved data quality, feature extraction methods, or alternative models like deep learning.

Result from step 2:

The SIFT method detected **73,976** total keypoints, of which **12,230** contributed to correct predictions, yielding a keypoint matching accuracy of **16.53%**. The ORB method detected a

larger number of keypoints (**138,939**) but achieved a lower matching accuracy of **11.90%**, with only **16,529** keypoints contributing to correct predictions. These results indicate that SIFT keypoints were more effective at aiding correct predictions despite ORB detecting more overall keypoints. The relatively low accuracy for both methods reflect the challenges of using these approaches for this classification task.

Result from step 3:

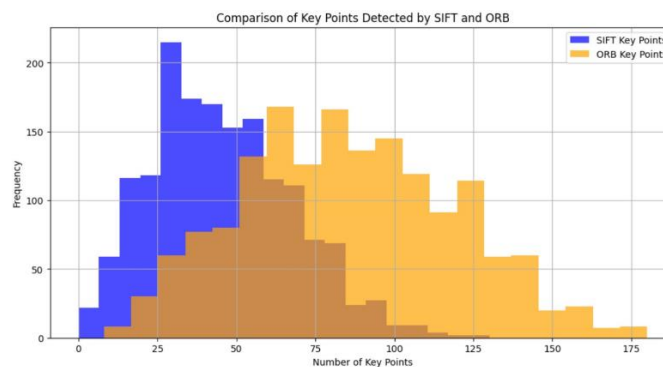
The SIFT method achieved a classification accuracy of **14.98%** with a total execution time of **16.48** seconds, while the ORB method achieved a lower accuracy of **12.71%** but with a significantly faster execution time of **7.46** seconds. These results demonstrate a trade-off between accuracy and computational efficiency: SIFT is more accurate but slower, whereas ORB is faster but less accurate. This insight is crucial for applications where the choice between precision and speed depends on the specific requirements of the task.

Result from step 4:

The robustness testing revealed that both SIFT and ORB are sensitive to certain transformations. SIFT showed better accuracy in most cases, such as rotation (**13.87%**) and negative brightness adjustment (**15.22%**), while ORB had higher performance in scaling up (**12.71%**). However, both methods struggled significantly with noise, with SIFT achieving only 1.53% accuracy and ORB **5.95%**. Overall, SIFT demonstrated higher robustness across a broader range of transformations, but ORB was more consistent in handling simpler cases like scaling and neutral brightness. These results highlight the need for adaptation of feature extraction techniques based on expected image conditions.

Result from step 5:

The results show that SIFT detected an average of **45.41 key points** per image, while ORB detected a significantly higher average of **85.29 key points** per image. The histogram reveals that SIFT tends to detect fewer key points, with most images having key points below 50, resulting in a left-skewed distribution. In contrast, ORB shows a wider spread and higher frequency of key points, with most values ranging between **50 and 150**. This indicates that ORB detects more key points overall compared to SIFT.



****Note that all output of each step is in the google Colab file (code file that we submitted)**

Performance of the algorithms

The performance of the SIFT and ORB algorithms reveals distinct characteristics based on the experimental results. SIFT, although highly accurate in detecting key points and providing detailed descriptors, shows relatively longer execution times compared to ORB. This makes SIFT less suitable for real-time applications where speed is a critical factor. In terms of accuracy, SIFT tends to perform better in terms of feature matching and classification under less challenging conditions. On the other hand, ORB offers a faster alternative with a tradeoff in accuracy. Despite its lower key point detection count and slightly reduced robustness, ORB maintains competitive performance, especially in real-time scenarios where computational efficiency is prioritized. Furthermore, both algorithms demonstrate vulnerability to transformations such as noise, scaling, and brightness changes, with SIFT generally outperforming ORB in robustness, though the difference in accuracy becomes more pronounced under extreme conditions. Overall, the choice between SIFT and ORB depends on the specific application requirements, balancing the need for speed versus robustness and accuracy.

Conclusion

In conclusion, this assignment provided a comprehensive comparison of the **SIFT** and **ORB** feature extraction algorithms, examining their performance in key point detection, feature matching, and robustness against image transformations. Through experimentation, we observed that **SIFT** excels in accuracy and robustness, making it ideal for tasks requiring precise key point matching, despite its higher computational cost. **ORB**, on the other hand, offers a faster and more efficient alternative with slightly reduced performance, making it suitable for real-time applications where speed is a priority. Additionally, both algorithms showed some vulnerability to transformations such as scaling, rotation, brightness adjustments, and noise, with **SIFT** generally maintaining better accuracy under these conditions. Ultimately, the choice between **SIFT** and **ORB** depends on the specific needs of the application, where **SIFT** is favored for its robustness and accuracy, and **ORB** is chosen for its speed and efficiency in resource-constrained environments.